

**Course Title: Development Geography
(GeES 3011)**



CHAPTER ONE

INTRODUCTION

1.1. Definition of Development Geography

- **Development geography** is a branch of geography which refers to the standard of living and its quality of life of its human inhabitants. In this context, development is a process of change that affects peoples' lives.
 - In development geography, geographers study spatial patterns in development. They try to find by what characteristics they can measure development by looking at economic, political and social factors. They seek to understand both the geographical causes and consequences of varying development.
 - Studies compare More Economically Developed Countries (MEDCs) with Less Economically Developed Countries (LEDCs). Additionally variations within countries are looked
 - Therefore, development geography studies about spatial variations of development and how geography/location of a place or country determines its development.
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1.2. The Concept of Development

❖ Traditional Definition - Economic Measures

- In strictly economic terms, development has traditionally meant **achieving sustained rates of growth of GDP and income per capita** to enable a nation to **expand its output** at a rate faster than the growth rate of its population.
- Levels and rates of growth of “real” per capita gross national income (GNI) (monetary growth of GNI per capita minus the rate of inflation) are then used to measure the overall economic well-being of a population—how much of **real goods and services** is available to the average citizen for consumption and investment.
- Economic development in the past has also been typically seen in terms of the **planned alteration of the structure of production and employment** so that agriculture’s share of both declines and that of the manufacturing and service industries increases.
- Problems of **poverty, discrimination, unemployment, and income distribution** were of **secondary** importance to “getting the growth job done.”



❖ Modern Definition of Development – Broad Measures

- For almost every writer a different definition of development exists.
- Rubenstein defines development as "the process of improving the material conditions of people through diffusion of knowledge and technology". If people have the know-how and the facilities to improve their lives, they are likely to do so. However, many scholars suggest that because of the world's political and economic structures, this diffusion may not be enough.
- According to Todaro, Development is not purely an economic phenomenon but rather a multi-dimensional process involving reorganization and reorientation of entire economic and social system.
- Development is the process of improving the entire living conditions of all people of a nation. Development is both a physical reality and state of mind in which societies have secured for obtaining better life. This concept is broader than the concept of economic growth.



- The current approach to development owes a great deal to the writings and views of Noble Laureate **Dr. Amartya Sen**. As Sen put it, ``**Economic growth cannot be sensibly treated as an end in itself. Development has to be more concerned with enhancing the lives we lead and the freedoms we enjoy.**'' This new approach is popularly known as Sen's **Capabilities Approach**.
- According to this approach, development is not just about increasing the availability of commodities (focus of the per-capita income approach) but expanding the capabilities of individuals to use these commodities and enhancing the freedom of choice of people. **Higher income is important, an element of one's well being. But, well being of individuals also depends on their health, education, geographical and social environment, and political system.**
- Thus, **Development is a process of improving the quality of all human lives with three equally important aspects: sustenance, self-esteem, and freedom from servitude.**

- **The three core values of development:** (i) sustenance, (ii) self-esteem, and (iii) freedom.

I. Sustenance - The Ability to Meet Basic Needs

- All people have certain basic needs without which life would be impossible. These life sustaining basic human needs include food, shelter, health, and protection. When any one of these is absent, or in critically short supply, a condition of “absolute underdevelopment” exists. The basic function of economic activities is to meet these basic needs.
- Thus, it can be claimed that economic development is a necessary condition for the improvement in the quality of life. Human beings are born with certain potential capacities. Without these basic needs, the utilization of human potential would not be possible.

II. Self-Esteem - to be a person

- Self-esteem is a **sense of worth and self-respect, not being used as a tool by others for its own ends**. All people and societies seek some basic form of self-esteem. Though they may call it **authenticity, identity, dignity, respect, honor** or **recognition** the nature and form of self-esteem may vary from society to society and from culture to culture.
- Of course, it is closely **associated with material prosperity**. Consequently, it is often difficult for those who are materially deprived to **experience sense of pride or self-esteem**

III. Freedom from Servitude – to be able to choose

- Freedom is understood in the sense of emancipation from alienating **material conditions of life** and from **social servitude to nature, ignorance, other people, misery, institutions, and dogmatic beliefs**, especially one's poverty is one's destination. **Freedom involves expanded range of choices for societies.**
- According to (Lewis, 2003), “the advantage of economic growth **is not that wealth increases happiness**, but that **it increases the range of human choice.**” Wealth can enable people to gain greater control over nature and the physical environment than they would have if they remained poor.
- The concept of **human freedom should also encompass various components of political freedom including but not limited to freedom of expression, personal security, political participation, and equality of opportunity.**

Todaro's Three Objectives of Development

- ❑ We may conclude that development is both a physical reality and a state of mind in which society has, through some combination of social, economic, and institutional processes, secured the means for obtaining a better life. Whatever the specific components of this better life, development in all societies must have at least the following three objectives:
1. To increase the availability and widen the distribution of basic life-sustaining goods such as food, shelter, health, and protection;
 2. To raise levels of living, including, in addition to higher incomes, the provision of more jobs, better education, and greater attention to cultural and human values, all of which will serve not only to enhance material wellbeing but also to generate greater individual and national self-esteem;
 3. To expand the range of economic and social choices available to individuals and nations by freeing them from servitude and dependence, not only in relation to other people and nation-states, but also to the forces of ignorance and human misery.

1.3. Indicators of Development

A. Economic Indicators

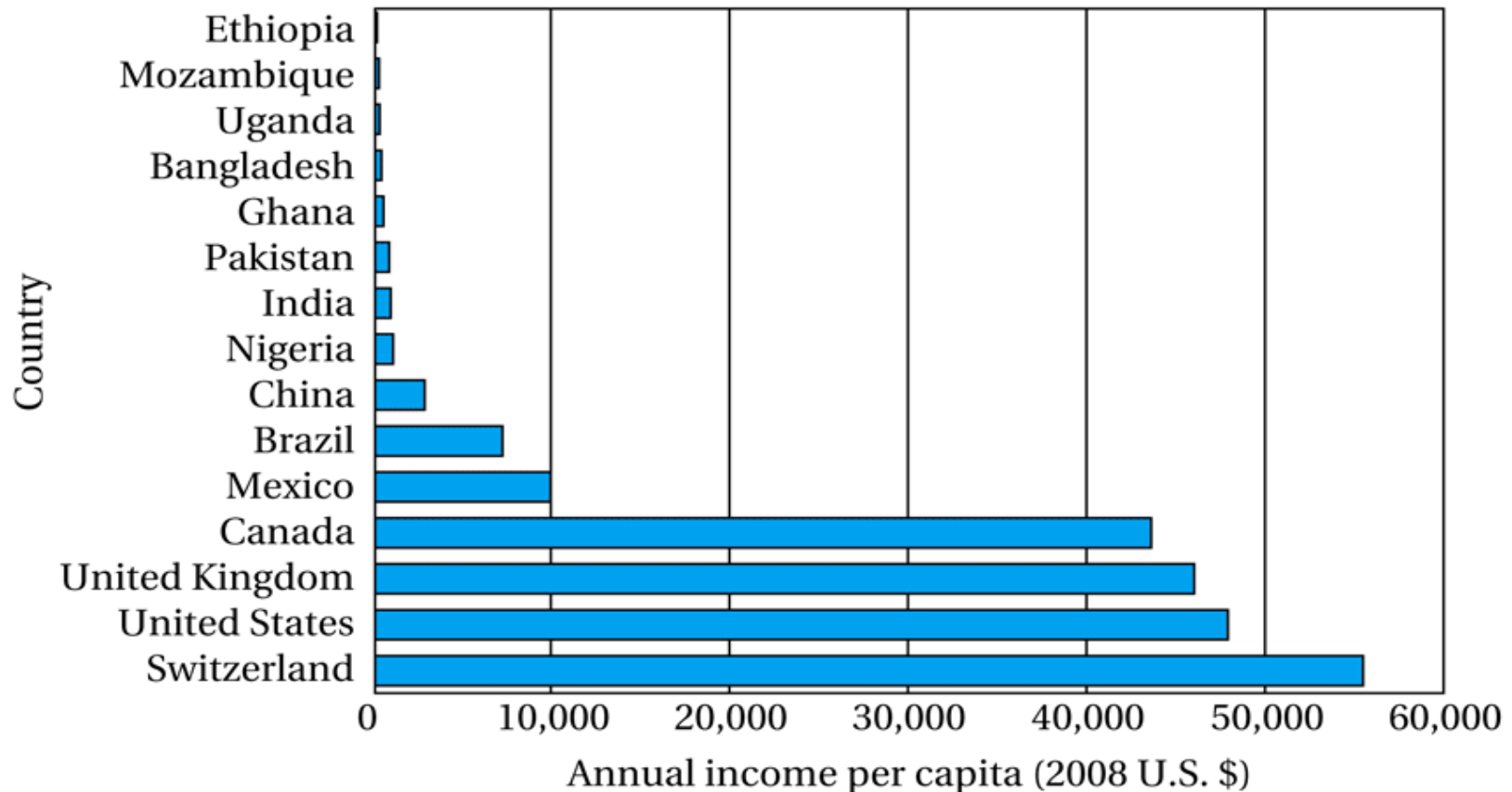
- **Per capita income (PCI)** is commonly used as an indicator of development. However, in many countries, especially the less developed countries, income figures are difficult to ascertain, so accuracy is always in question.
- **The gross domestic product (GDP)**-the monetary value of the total goods and services produced in a country annually-is often used as an alternative to PCI. However, this number must be standardized, because a large country like the United States will obviously have a larger GDP than a small country like Djibouti.
- So the GDP is divided by the total population in a country to create **per capita GDP**, which can be compared for different countries. However, per capita GDP (nominal) gives no indication of income disparity within a given country because it is an average.
- Generally, MDCs have higher GDP and Per capita income than LDCs (though the later one depends on number of population of the country)



- The **economic structure** of a given region, or the types of jobs available in a region, is another widely-used development indicator. Typically, LDCs will have **a higher number of jobs in the primary and secondary sectors**, and MDCs often have a **higher percentage of the workforce employed in the tertiary sector**.
- A third economic indicator of development is **worker productivity**. Because technology is more prevalent in MDCs, worker productivity is generally higher in these countries.



Figure 1.1 Income Per Capita in Selected Countries (2008)



Source: Data from World Bank, *World Development Indicators*, 2010 (Washington, D.C.: World Bank, 2010), tab. 1.1.

Table 1.1 A Comparison of Per Capita GNI, 2008

Country	GNI Per Capita (U.S. \$)	
	Exchange Rate	Purchasing Power Parity
Argentina	7,190	13,990
Bangladesh	520	1,450
Brazil	7,300	10,070
Burundi	140	380
Cameroon	1,150	2,170
Chile	9,370	13,240
China	2,940	6,010
Costa Rica	6,060	10,950
Ghana	630	1,320
Guatemala	2,680	4,690
India	1,040	2,930
Indonesia	1,880	3,590
Kenya	730	1,550
Malawi	280	810
Malaysia	7,250	13,730
Mexico	9,990	14,340
Nicaragua	1,080	2,620
Sierra Leone	320	770
South Korea	21,530	27,840
Sri Lanka	1,780	4,460
Thailand	3,670	7,760
Uganda	420	1,140
United Kingdom	46,040	36,240
United States	47,930	48,430
Venezuela	9,230	12,840
Zambia	950	1,230

Source: Data from World Bank, *World Development Indicators, 2010* (Washington, D.C.: World Bank, 2010) tab. 1.1.

B. Social Indicators

- Social indicators also provide a way to measure development.
- The first social indicator has to do with **education and literacy**. Often, there are great discrepancies between MDCs and LDCs in these categories.
- In many countries, **education levels for males and females vary widely**. Today many geographers use the levels of female education and literacy to measure development, since female literacy rates in MDCs are much higher than in LDCs. Often there are cultural reasons for this discrepancy, such as in Afghanistan where female education was severely curtailed under Taliban rule.
- **Health and welfare services** are also more extensive in more developed countries, mostly because additional resources can be allocated to these services. Not only do these resources assist sick or destitute people, but they also help prevent diseases.
- Examples of the extent of such services include **the number of doctors per 10,000 people in a country**, the **degree of access to clean water and sanitation**, the **level of available housing**, and **average caloric intake**. These social indicators are frequently used to measure the living standards of people.



Table 1.2 Primary School Enrollment and Pupil-Teacher Ratios, 2010

Region or Group	Net Primary School Enrollment (%)	Primary Pupil-Teacher Ratio
Income Group		
Low	80	45
Lower Middle	87	23 ^a
Upper Middle	94	22
High	95	15
Region		
East Asia and Pacific	93 ^a	19
Latin America and the Caribbean	94	25
Middle East and North Africa	91	24
South Asia	86	40 ^a
Sub-Saharan Africa	73	49
Europe and Central Asia	92	16

Source: Data from World Bank, *World Development Indicators, 2010* (Washington, D.C.: World Bank, 2010), tabs 2.11 and 2.12.

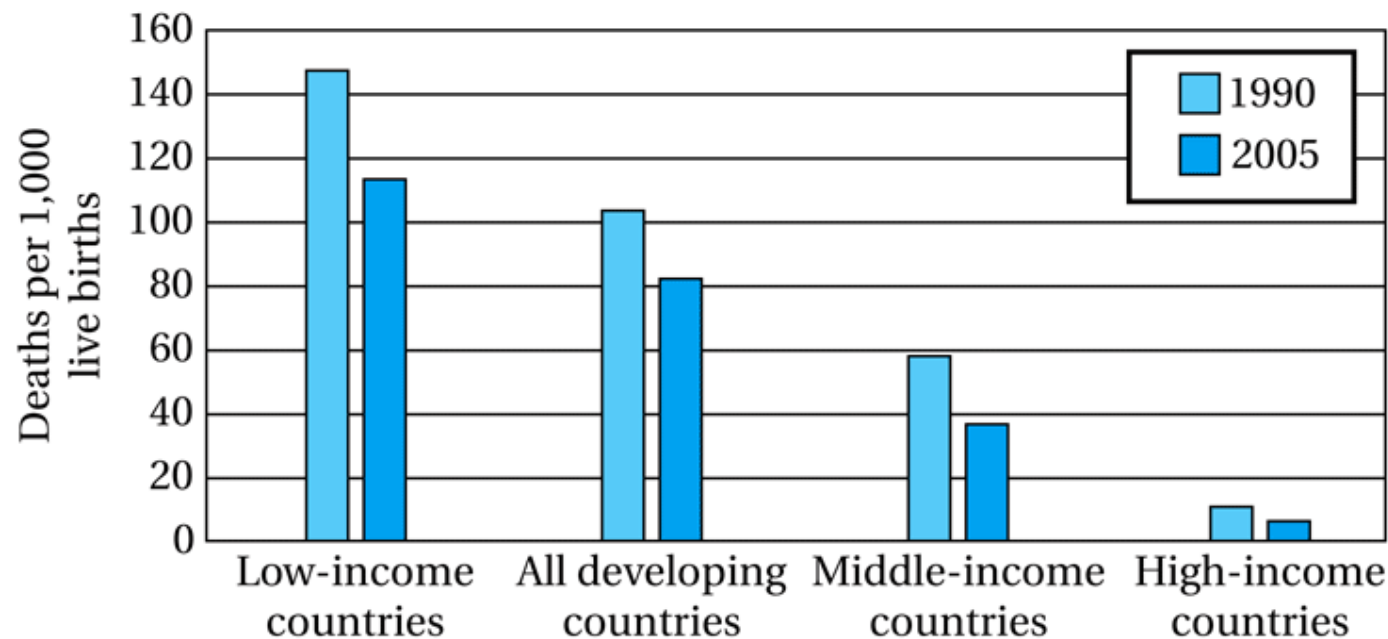
^aData for 2009

C. Demographic Indicators

- Demographic indicators comprise the third set of most frequently used indicators of development.
- These indicators include infant mortality rates, natural increase rates, crude birth rates, and life expectancy.
- However, in order to measure development, they are much more informative when used in comparisons with other countries.



Figure 1.2 Under-5 Mortality Rates, 1990 and 2005



Source: International Bank for Reconstruction and Development/World Bank, *World Development Indicators, 2007* (Washington, D.C.: World Bank, 2007), p. 36. Reprinted with permission.

- The United Nations has designed a method for measuring the development level of a country by combining several indicators. It is called the **Human Development Index (HDI)**.
- The HDI uses **per capita GDP**, **literacy rate**, **amount of education**, and **life expectancy** to create an index that ranges from (0 – 1), 1.0 representing the highest level of development, while 0 representing the lowest level. This index is very useful as we begin to recognize varying levels of development around the world.



Table 1.3 Commonality and Diversity: Some Basic Indicators

Country or Group	2008 Income Per Capita (U.S. \$)	2008 PPP Per Capita (U.S. \$)	2008 Life Expectancy (years)	Prevalence of Undernourish- ment ^a (%)	2007 Under-5 Mortality per 1,000 Live Births	2008 Crude Birth Rate	Adult Literacy ^b	
							Male	Female
Income Group								
Low	523	1,354	59	30	118	32	76	63
Lower middle	2,073	4,589	68	15	64	20	87	73
Upper middle	7,852	12,208	71	6	23	17	95	92
High	39,687	37,665	80	5	7	12		
Country								
Dem. Rep. Congo (LIC)	150	280	48	75	199	45	78	56
India (LMC)	1,040	2,930	64	22	69	23	75	51
Egypt (LMC)	1,800	5,470	70	<5	25	25	75	58
Brazil (UMC)	7,300	10,070	72	6	22	16	90	90
Malaysia (UMC)	7,250	13,730	74	<5	6	20	99	94
United States (high-income)	47,930	48,430	78	<5	8	14		
Region								
East Asia and the Pacific	2,644	10,461	72	12	29	14	96	90
Latin America and the Caribbean	6,768	10,312	73	9	23	19	92	91
Middle East and North Africa	3,237	7,343	71	7	34	24	82	65
South Asia	963	2,695	64	22	76	24	73	50
Sub-Saharan Africa	1,077	1,949	52	28	144	38	74	57
Europe and Central Asia	7,350	11,953	70	6	22	14	99	97

Source: Data from World Bank, *World Development Indicators*, 2010 (Washington, D.C.: World Bank, 2010), multiple tables.

^aMost recent year between 2004 and 2006.

^bMost recent year between 2005 and 2008.

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1.4. Growth and Development

1.4.1 Economic Growth

- The term economic growth is defined as **the process whereby the country's real national and per capita income increases over a long period of time**. It refers to the **increase in the total wealth of a nation regardless of population**. It is used to **refer increasing output in production (GDP or GNP)**. This definition of economic growth consists of the following **features of economic growth**:
- Economic Growth implies a **process of increase in National Income and Per-Capita Income**. The increase in Per-Capita income is the better measure of Economic Growth since it reflects increase in the improvement of living standards of masses.
- Economic Growth is **measured by increase in real National Income and not just the increase in money income or the nominal national income**. In other words the increase should be in terms of **increase of output of goods and services**, and not due to a mere increase in the market prices of existing goods.



- **Increase in Real Income should be Over a Long Period:** The increase of real national income and per-capita income should be sustained over a long period of time. The short-run seasonal or temporary increases in income should not be confused with economic growth.
- **Increase in income should be based on Increase in Productive Capacity:** Increase in Income can be sustained only when this increase results from some **durable increase in productive capacity** of the economy like **modernization** or use of **new technology in production**, **strengthening of infrastructure** like **transport network**, improved **electricity** generation etc.



1.4.2 Economic Development

- Economic development is defined as **a sustained improvement in material well being of society**.
- Economic development is a **wider concept than economic growth**.
- Apart from growth of **national income**, it **includes changes – social, cultural, political as well as economic** which contribute to material progress.
- It contains **changes in resource supplies, in the rate of capital formation, in size and composition of population, in technology, skills and efficiency, in institutional and organizational set-up**. These changes fulfill the wider objectives of ensuring more equitable income distribution, greater employment and poverty alleviation.
- In short, economic development is a process consisting of **a long chain of interrelated changes in fundamental factors** of supply and in the structure of demand, leading to a rise in the net national product of a country in the long run.



- The **economic growth** is a narrow term. It involves increase in output in quantitative terms but **economic development includes changes in qualitative terms** such as social attitudes and customs along with quantitative growth of output or national income.
- **Economic development without growth is almost inconceivable.**
- **Economic development is the process by which simple, low income national economies are transformed into modern industrial economies. It involves a rise in real income per person. Therefore, economic development comprises both the emergence of sustained and rapid extensive growth, and the emergence of sustained and rapid intensive growth.**
- The comparison between the two concepts is given in the following table:



	Economic Growth	Economic Development
Meaning	Economic growth refers to an increase in the real output of goods and services in the country.	Economic development implies changes in income, savings and investment along with progressive changes in socio-economic structure of country (institutional and technological changes)
Factors:	Growth relates to a gradual increase in one of the components of Gross Domestic Product: consumption, government spending, investment, net exports	Development relates to growth of human capital , decrease in inequality figures, and structural changes that improve the quality of life of the population
Measurement:	Economic Growth is measured by quantitative factors such as increase in real GDP or per capita income (eg 3% growth)	The qualitative measures such as HDI (Human Development Index), gender-related index , Human poverty index (HPI), infant mortality, literacy rate etc. are used to measure economic development.
Effect:	Economic growth brings quantitative changes in the economy.	Economic Development leads to qualitative as well as quantitative changes in the economy.
Relevance:	Economic growth reflects the growth of national or per capita income.	Economic development reflects progress in the quality of life in a country

Economic Growth (GDP)

Increases in Material Output



Economic Development

Greater emphasis on social objectives
e.g. poverty, education, health,



Sustainable Economic Development

Recognising the need for environmental
Protection, inter- and intra-generational
equity.

1.5. Space and Development

- There is a considerable **spatial variation** in development rates.
- The most famous pattern in development is the **North-South divide**. The North-South divide separates **the rich North or the developed world**, from **the poor South**. This line of division is not as straightforward as it sounds and splits the globe into two main parts. It is also known as **the Brandt Line**.
- The "**North**" in this divide is regarded as being **North America, Europe, Russia, South Korea, Japan, Australia, New Zealand** and the like. The countries within this area are generally the **more economically developed**.
- The "South" therefore encompasses the remainder of the Southern Hemisphere.
- Another possible dividing line is the **Tropic of Cancer** with the **exceptions of Australia and New Zealand**.
- It is critical to understand that the status of countries is far from static and the pattern is likely to become distorted with the fast development of certain southern countries, many of them **NICs** (Newly Industrialized Countries) including **India, Thailand, Brazil, Malaysia, Mexico** and others. These countries are experiencing sustained fast development on the back of growing manufacturing industries and exports.



- Many African nations have recently experienced **reduced GNPs due to wars and the AIDS epidemic**, including Angola, Congo, Sierra Leone and others.
- Arab **oil producers** rely very heavily on **oil exports** to support their GDPs so any reduction in oil's market price can lead to rapid decreases in GNP.
- **Countries which rely on only a few exports** for much of their income are very **vulnerable to changes in the market value of those commodities** and are often derogatively called **banana republics**.
- **Many developing countries do rely on exports of a few primary goods** for a large amount of their income (**coffee** and **timber** for example), and this can create havoc when the value of these commodities drops, leaving these countries with no way to pay off their debts.
- Within countries the pattern is that **wealth is more concentrated around urban areas than rural areas**. Wealth also tends towards areas with **natural resources or in areas that are involved in tertiary (service) industries and trade**. This leads to a gathering of wealth around mines and monetary centers such as New York, London and Tokyo.

- Geography can also affect economic development in a number of ways.
- First, access to sea routes is important; this has been noted as far back as Adam Smith. Sea travel is much cheaper than that of land, leading to a wider dissemination of both resources and ideas, both of which are integral to economic stimulus.
- Geography also dictates the prevalence of disease: for example, Malaria is largely associated with nations that have struggled to achieve sound economic development. Not only does disease decrease labor productivity, but it changes the age structure of the country, forcing the population to lean heavily toward children as adults die from disease and the population sees an increase of fertility to keep up with the high death rates.
- The third way geography affects development is through agricultural productivity. **Temperate regions** have shown the highest output of major grains; regions such as the **African savanna** relatively yield much less value for the labor cost. Low agricultural output means that a larger portion of the population must spend their efforts in agriculture, leading to a slower urban development.



- Generally, geographic location of a country or place, together with other factors, could determine the country's development as:
 1. **Absolute location** of the country determines its potential for development depending on the country's endowments including its climate, soil type, minerals, vegetation....
 2. **Relative location** of a country or place, including access to sea routes, political stability of the region, and proximity to a well developed region, ... determines the country's opportunity for development.



1.6. Classification of Countries and their Defining Characteristics

- While some writers divide countries of the world into **more developed countries (MDCs)** and **less developed countries (LDCs)**, some geographers define these countries differently.
- Based on income countries can also be divided into three: **low income**, **middle income** and **high income** countries.
- Since 1940s the world had also been divided into **first-world (Capitalist)**, **second-world (Socialist)**, and **third-world countries** according to the geopolitics of the Cold War. The poorer countries of the world were called the third world. Today, most geographers discount this term as being a relict of the Cold War and consider it condescending.



- Another popular distinction is what is termed the north-south split. The line running at approximately 30° north latitude separates the world into MDCs and LDCs (with the notable exception of Australia and New Zealand).
- This distinction is important in that it shows the geographic distribution of countries at the different levels of development. As there are fewer negative connotations with using the terms north and south, the terms MDC and LDC are more acceptable.



- The difference between the two categories should be easy to envision.
- MDCs, such as the United States and Japan, have **relatively high standards of living, high levels of educational attainment, long life expectancies**, and **low infant mortality rates**. Most residents of MDCs have **access to clean water, sewage, and electricity**.
- In contrast, many residents of LDCs **lack these basic conditions** that people in MDCs tend to take for granted. It is important to remember that these are very broad categorizations and much disparity exists within each category.



○ Common Features of Developing Countries

1. **Low per Capita Income:** The level of per capita income is very low in underdeveloped countries.
2. **Poor Level of Living:** The vast majority of people in underdeveloped nations lie under the conditions of **poverty**, **malnutrition**, **disease**, **illiteracy**, etc. Even basic necessities of life such as minimum **food**, **clothing** and **shelter** are not easily accessible to the poor masses.
3. **High Rate of Growth of Population:** Population growth in underdeveloped countries neutralizes economic growth. **High population implies greater consumption expenditure and lower investments in productive activities and slows down the economic development.**



4. **Highly Unequal Income Distribution:** The **income inequality** between the rich and the poor people within the underdeveloped countries is also very high.
5. **Prevalence of Mass Poverty:** Low level of per capita income combined with high degree of inequalities in its distribution leads to widespread poverty in underdeveloped countries.
6. **Low Levels of Productivity:** The Productivity level (i.e. **output produced per person**) tends to be very low in an underdeveloped country which is mainly due to :
 - i. **inefficient workforce** which itself is a consequence of poverty, ill health and lack of education
 - ii. **Low work culture (ethics)**
 - iii. **Low use of capital** in the form of machinery and equipment.



7. **Low Rate of Capital Formation:** The **saving rate** in an underdeveloped country is quite low and rate of capital formation also is very slow.
8. **Technological Backwardness:** In most of the sectors of an underdeveloped economy the **techniques of production employed are generally obsolete**, mainly due to low saving rate.
9. **High Level of Unemployment:** Unemployment levels are very high in the underdeveloped countries mainly due to **lack of capital** and **low level of development** in various economic sectors, these countries are not able to absorb the rising labor supply.
10. **Low Social Indicators of Development:** The under-developed countries have very low social indicators such as **low literacy rate**, **high infant mortality rate**, **low expectancy of life**, etc. as compared to the developed countries.



Other Common Characteristics of Developing Countries

- These features in common are *on average and with great diversity*, in comparison with developed countries:
 - Lower levels of living and productivity
 - Lower levels of human capital
 - Higher levels of inequality and absolute poverty
 - Higher population growth rates
 - Greater social fractionalization
 - Larger rural population - rapid migration to cities
 - Lower levels of industrialization and manufactured exports
 - Adverse geography
 - Underdeveloped financial and other markets (Imperfect market, incomplete information)
 - Colonial Legacies - poor institutions etc.

1.7. Development Problems: Poverty and Inequality

- Although there are numerous problems in developing countries, **poverty and inequality are the most prominent ones.**
- Many developing countries that had experienced relatively high rates of economic growth by historical standards discovered that such growth often **brought little in the way of significant benefits to their poor.**
- Because the **elimination of widespread poverty and high and even growing income inequality are at the core of all development problems** and in fact define for many people the principal objective of development policy.
- Although our main focus is on **economic poverty and inequalities in the distribution of incomes and assets**, it is important to keep in mind that this is only part **of the broader inequality problem in the developing world.**



- Of equal or even greater importance are **inequalities of power, prestige, status, gender, job satisfaction, conditions of work, degree of participation, freedom of choice**, and many other dimensions of the problem that relate more to our second and third components of the meaning of development, self-esteem, and freedom to choose.
- As in most social relationships, **we cannot really separate the economic from the noneconomic manifestations of inequality**. Each reinforces the other in a complex and often interrelated process of cause and effect.



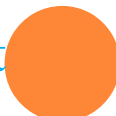
1.7.1. Poverty

- According to (The World Bank, 2011) **poverty is defined as “... a pronounced deprivation in wellbeing.”** This of course requests the questions of what is meant by well-being and of what is the reference point against which to measure deprivation.
- **There are different approaches to treat well-being.**
- One approach is to think of well-being as the **command over commodities in general**, so people are better off if **they have a greater command over resources**.
 - The main focus is on whether households or individuals have enough **resources to meet their needs**.
 - Typically, poverty is then measured by comparing individuals' income or consumption with some defined threshold below which they are considered to be poor.
 - This is the most conventional view—**poverty is seen largely in monetary terms**—and is the starting point for most analyses of poverty.



- A second approach to well-being (& hence poverty) is to ask whether people are able to obtain a specific type of **consumption good**:
 - Do they have enough food?
 - Or shelter?
 - Or health care?
 - Or education?
- In this view the analyst **goes beyond the more traditional monetary measures** of poverty:
 - Nutritional poverty might be measured by examining whether children are **stunted or wasted**; or whether people are consuming less than a reasonable minimum calories - often set at **2,100 Calories per person per day**; and
 - educational poverty might be measured by asking whether people are **literate** or **how much formal schooling they have received**.



- According to (The World Bank, 2011) Perhaps the broadest approach to well-being is the one articulated by **Amartya Sen (1987)**, who **argues that well-being comes from a capability to function in society**. Thus, **poverty arises when people lack key capabilities**, and so have either some or all of the following incapacibilities:
 - inadequate income or
 - education, or
 - poor health, or
 - insecurity, or
 - low self-confidence, or
 - a sense of powerlessness, or
 - the absence of rights such as freedom of speech
 - When viewed in this way, **poverty is a multidimensional phenomenon** and less amenable to simple solutions. **For instance, while higher average incomes will certainly help reduce poverty, these may need to be accompanied by measures to empower the poor, or insure them against risks, or to address specific weaknesses such as inadequate availability of schools or a corrupt health service.**
- 

- Furthermore, the United Nations, further defines absolute poverty as the absence of **any two** of the following eight basic needs:
 - **Food:** Body Mass Index must be above **8**.
 - **Safe drinking water:** Water must not come from solely rivers and ponds, and must be available nearby (**less than 15 minutes' walk each way**).
 - **Sanitation facilities:** Toilets or latrines must be accessible in or near the home.
 - **Health:** Treatment must be received for serious illnesses and pregnancy.
 - **Shelter:** **Homes must have fewer than four people living in each room.** Floors must not be made of **dirt, mud**, or clay.
 - **Education:** Everyone must attend school or otherwise learn to read.
 - **Information:** Everyone must have **access to newspapers, radios, televisions, computers, or telephones at home**.
 - **Access to services:** indicates the complete availability of **education, health, legal, social, and financial** (credit) services.



- **Poverty Line** – is the **threshold of poverty below which one's income does not cover basic necessities**. In most cases the poverty line is taken as a line representing an income of **1 dollar a day or 2 dollar a day**. People whose daily income is below the poverty line are said to be in absolute poverty.
- There are two types of poverty
- **Relative Poverty** – is a condition which shows that some people are poorer compared to other people. Relative poverty exists in all societies, as income and wealth of all people can not be equal.
 - In most cases, few rich individuals possess the vast majority of wealth and income of the country, while the remaining population possess only small percentage. Eg. The rich who make only 20% of the population may own 80% of the wealth in a country.
- **Absolute Poverty** - The situation of being unable or only barely able to meet the subsistence essentials of **food, clothing, and shelter**. People in absolute poverty are counted as the total number living below a specified minimum level of real income—(those whose daily income is below an international poverty line).
 - Absolute poverty is common in developing countries.
 - It shows percentage of people who earn below the poverty line.



1.7.2 Inequality

- Inequality is a **broader concept than poverty** in that it is defined over the entire population, not just for the portion of the population below a certain poverty line.
- Most inequality measures do not depend on the mean of the distribution; this property of mean independence is considered to be a desirable feature of an inequality measure.
- Of course, **inequality measures are often calculated for distributions** rather than expenditure - for instance, **for income, land, assets, tax payments**, and many other continuous and key variables.
- Although there is inequality in every society, it is **more pronounced in developing countries**. Eg. **Income inequality, gender inequality, political inequality, social inequality** etc.



WHY DOES INCOME INEQUALITY EXIST?

- ❑ Because people do not receive the same **labor income, asset income, and transfer payments, or pay the same taxes.**
- ❑ Difference in possessions of **natural resources** also results in income inequalities



FACTORS CONTRIBUTING TO INCOME INEQUALITY

- **Innate Abilities and Attributes:** Individuals are not all born with the same abilities and qualities.
- **Work and Leisure:** There is a tradeoff between work and leisure: More work means less leisure, less work means more leisure.
- **Education and Other Training:** Generally, this is human capital – the education, the development of skills, and anything else that is particular to the individual and increases his or her productivity.



- **Risk Taking:** Individuals have different attitude towards risk.
- **Luck:** When individuals can't explain why something has happened to them, they often say it was the result of good or bad luck.
- In the long run, such factors as innate ability and attributes, education, and personal decisions are more likely to have a larger sustained effect on income than good or bad luck.
- **Wage Discrimination:** This exists when individuals of equal ability and productivity, as measured by their marginal revenue products, are paid different wage rates by the same employer.



Measuring Inequality

- Inequality measures are often calculated for **distributions** rather than expenditure. For instance, inequalities of **income**, **land**, **assets**, **tax payments**, and many other continuous and key variables of inequality among individuals are measured on the basis of difference in their distribution.
- There are different methods of measuring inequality, particularly **income inequality**. The following are some of them:
 - Size distributions (quintiles, deciles)
 - Kuznets ratio
 - Lorenz curves
 - Gini coefficients and aggregate measures of inequality
 - Functional distributions [The distribution of income to factors of production (land, labor, and capital)]

a) Size distributions (quintiles, deciles)

- The personal or size distribution of income is the measure most commonly used by economists. It simply deals with individual persons or households and the total incomes they receive.
- Economists and statisticians therefore like to arrange all individuals by ascending personal incomes and then divide the total population into distinct groups, or sizes.
- A common method is to arrange all individuals according to ascending income levels, and then divide the population into successive quintiles (fifths) (dividing the population in to 5 groups, i.e each group representing 20% of the total), or deciles (tenths) (dividing in to 10 groups, each group representing 10% of the total). This helps to determine what proportion of the total national income is received by each income group (decile or quintile).

Table 2.1 Typical Size Distribution of Personal Income in a Developing Country by Income Shares—Quintiles and Deciles

Individuals	Personal Income (money units)	Share of Total Income (%)	
		Quintiles	Deciles
1	0.8		
2	1.0		1.8
3	1.4		
4	1.8	5	3.2
5	1.9		
6	2.0		3.9
7	2.4		
8	2.7	9	5.1
9	2.8		
10	3.0		5.8
11	3.4		
12	3.8	13	7.2
13	4.2		
14	4.8		9.0
15	5.9		
16	7.1	22	13.0
17	10.5		
18	12.0		22.5
19	13.5		
20	15.0	51	28.5
Total (national income)	100.0	100	100.0

- For example, Table 2.1 shows a hypothetical but fairly typical distribution of income for a developing country. In this table, 20 individuals, representing the entire population of the country, are arranged in order of ascending annual personal income, ranging from the individual with the lowest income (0.8 units) to the one with the highest (15.0 units).
- The total or national income of all individuals amounts to 100 units and is the sum of all entries in column 2.
- In column 3, the population is grouped into **quintiles of 4 individuals each**. The **first quintile represents the bottom 20%** of the population on the income scale. This group receives only 5% (i.e., a total of 5 money units) of the total national income.
- The second quintile (individuals 5 through 8) receives 9% of the total income. Alternatively, the bottom 40% of the population (quintiles 1 plus 2) is receiving only 14% of the income, while the **top 20% (the fifth quintile) of the population receives 51%** of the total income.



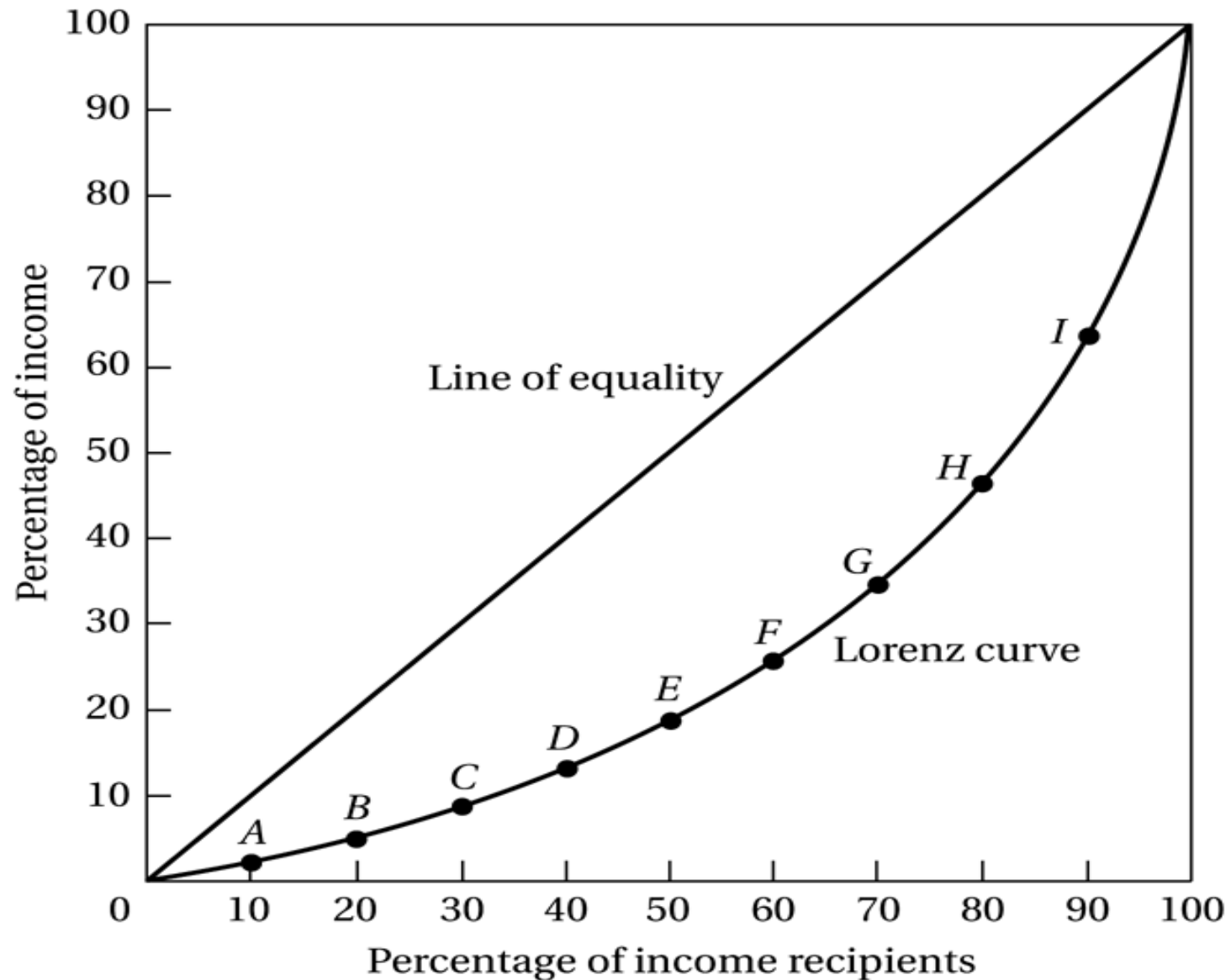
- A common measure of income inequality that can be derived from column 3 is the **ratio of the incomes received by the top 20% and bottom 40%** of the population. This ratio, sometimes called a **Kuznets ratio**, has often been used as a measure of the degree of inequality between high- and low-income groups in a country. In our example, this inequality ratio is equal to **51 divided by 14, or approximately 3.64**.
- To provide a more detailed breakdown of the size distribution of income, **decile (10%)** shares are listed in column 4. We see, for example, that the bottom 10% of the population (the two poorest individuals) receives only 1.8% of the total income, while the top 10% (the two richest individuals) receives 28.5%.
- Finally, if we wanted to know what the top 5% receives, we would divide the total population into 20 equal groups of individuals (in our example, this would simply be each of the 20 individuals) and calculate the percentage of total income received by the top group. In Table 2.1, we see that the top 5% of the population (the twentieth individual) receives 15% of the income, a higher share than the combined shares of the lowest 40%.



b) Lorenz Curves

- Another common way to analyze personal income statistics is to construct what is known as a Lorenz curve. Figure 2.1 shows how it is done.
- The **numbers of income recipients are plotted on the horizontal axis**, not in absolute terms but in **cumulative percentages**. For example, at point 20, we have the lowest (poorest) 20% of the population; at point 60, we have the bottom 60%; and at the end of the axis, all 100% of the population has been accounted for.
- The **vertical axis shows the share of total income received by each percentage of population**. It is also **cumulative up to 100%**, meaning that both axes are the same length.
- The entire figure is enclosed in a square, and a diagonal line is drawn from the lower left corner (the origin) of the square to the upper right corner.
- **At every point on that diagonal, the percentage of income received is exactly equal to the percentage of income recipients**—for example, the point halfway along the length of the diagonal represents 50% of the income being distributed to exactly 50% of the population. At the three-quarters point on the diagonal, 75% of the income would be distributed to 75% of the population.

Figure 2.1 The Lorenz Curve



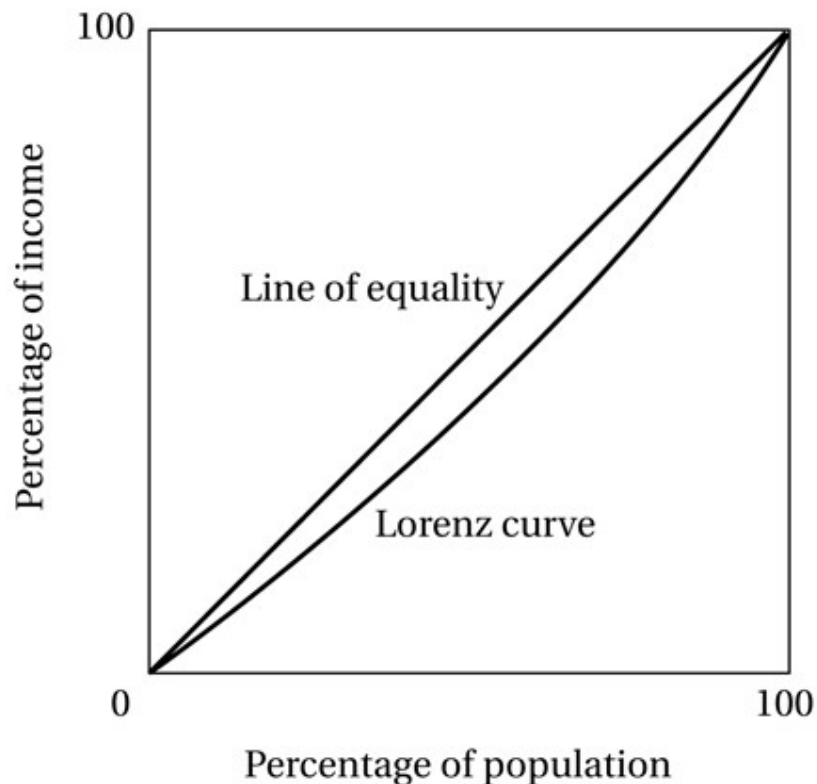
- In other words, the diagonal line in Figure 2.1 is representative of “**perfect equality**” in size distribution of income. Each percentage group of income recipients is receiving that same percentage of the total income; for example, the bottom 40% receives 40% of the income, while the top 5% receives only 5% of the total income.
- The **Lorenz curve shows the actual quantitative relationship** between the percentage of income recipients and the percentage of the total income they did in fact receive during, say, a given year.
- In Figure 2.1, we have plotted this Lorenz curve using the **decile data** contained in Table 2.1. In other words, we have divided both the horizontal and vertical axes into ten equal segments corresponding to each of the ten decile groups. Point A shows that the bottom 10% of the population receives only 1.8% of the total income, point B shows that the bottom 20% is receiving 5% of the total income, and so on for each of the other eight cumulative decile groups. Note that at the halfway point, 50% of the population is in fact receiving only 19.8% of the total income.



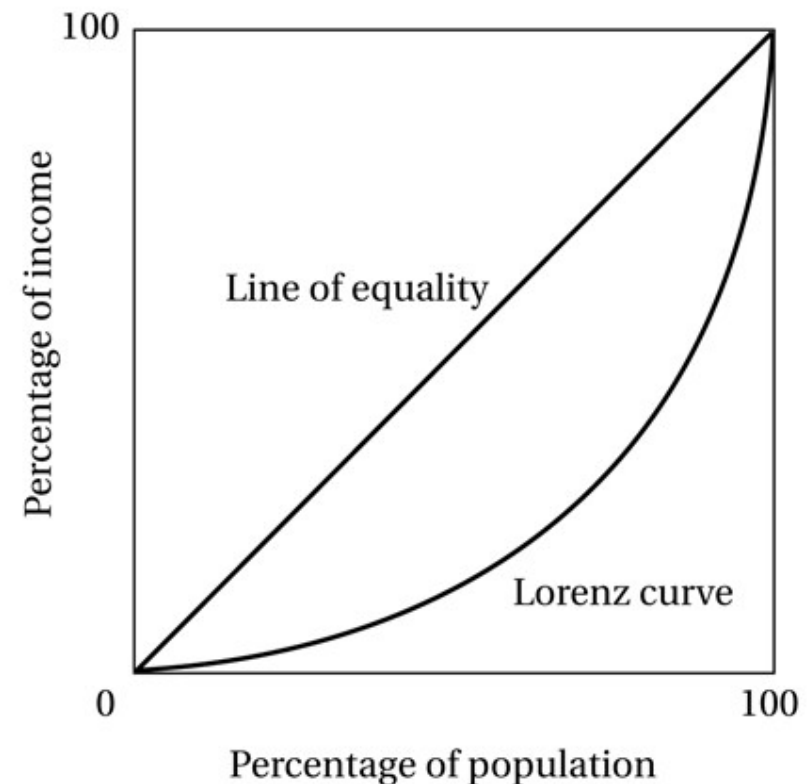
- The more the Lorenz curve is away from the diagonal (line of perfect equality), the greater the degree of inequality represented.
- The extreme case of **perfect inequality** (i.e., a situation in which one person receives all of the national income while everybody else receives nothing) would be represented by the congruence of the Lorenz curve with the bottom horizontal and right-hand vertical axes.
- Because no country exhibits either perfect equality or perfect inequality in its distribution of income, the Lorenz curves for different countries will lie somewhere to the right of the diagonal in Figure 2.1.
- The **greater the degree of inequality, the greater the bend** and the closer to the bottom horizontal axis the Lorenz curve will be.
- Two representative distributions are shown in Figure 5.2, one for a relatively equal distribution (Figure 5.2a) and the other for a relatively unequal distribution (Figure 5.2b).



Figure 5.2 The Greater the Curvature of the Lorenz Line, the Greater the Relative Degree of Inequality



(a) A relatively equal distribution



(b) A relatively unequal distribution